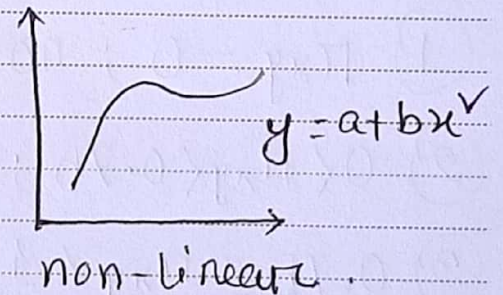
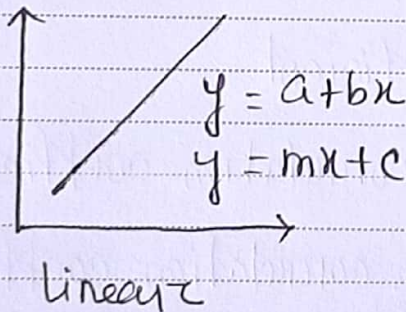




## Correlation coefficient & Regression analysis

\* Correlation :- Correlation measures the strength of linear relationship between two or more variables. Change in one variable effect to the other variable.

\* Correlation coefficient :- The numerical value by which the strength of linear relationship between two or more variables is measure.



\* Karl Pearson (1890) —

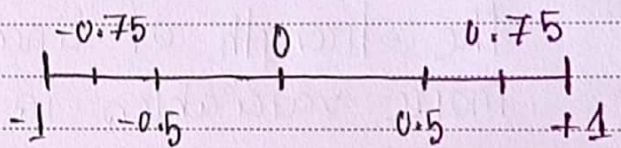
$$r_{xy} = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{n(\sum x^2) - (\sum x)^2} \sqrt{n(\sum y^2) - (\sum y)^2}}$$

⇒ Karl Pearson's coefficient of correlation / Pearson's product moment correlation.



\* Properties of Correlation coefficient :-

- ① Correlation coefficient lies between  $-1 < r_{xy} < +1$ .
- ② Correlation coefficient is symmetric i.e.  $r_{xy} = r_{yx}$ .
- ③ Correlation coefficient is the geometric mean of regression coefficients i.e.  $r_{xy} = \sqrt{b_{yx} \times b_{xy}}$
- ④ for two independent variable correlation coefficient is zero.
- ⑤ It is always unit free.

\* Interpret the value of  $r_{xy}$  :-

- ①  $r_{xy} = 0$  ; no correlation coefficient.
- ②  $0 < r_{xy} < 0.75$  ; simple positive correlation coefficient.
- ③  $0.75 \leq r_{xy} < 1$  ; strongly positive correlation coefficient.
- ④  $r_{xy} = +1$  ; Perfect positive correlation coefficient.
- ⑤  $-0.75 < r_{xy} < 0$  ; negative correlation coefficient.
- ⑥  $-1 < r_{xy} \leq -0.75$  ; strongly negative correlation coefficient.
- ⑦  $r_{xy} = -1$  ; Perfect negative correlation coefficient.

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Qna:Consider the following data -  $n = 6$ 

| $x$             | $y$             | $x^2$             | $y^2$              | $xy$             |
|-----------------|-----------------|-------------------|--------------------|------------------|
| 1               | 6               | 1                 | 36                 | 6                |
| 2               | 4               | 4                 | 16                 | 8                |
| 3               | 3               | 9                 | 9                  | 9                |
| 4               | 5               | 16                | 25                 | 20               |
| 5               | 4               | 25                | 16                 | 20               |
| 6               | 2               | 36                | 4                  | 12               |
| $\Sigma x = 21$ | $\Sigma y = 24$ | $\Sigma x^2 = 91$ | $\Sigma y^2 = 106$ | $\Sigma xy = 75$ |

① Compute Karl Pearson's correlation coefficient and comment.

② Draw scatter diagram.

Soln:

① We know that,

$$r_{xy} = \frac{n \Sigma xy - \Sigma x \Sigma y}{\sqrt{n(\Sigma x^2) - (\Sigma x)^2} \cdot \sqrt{n(\Sigma y^2) - (\Sigma y)^2}}$$

$$= -0.68$$

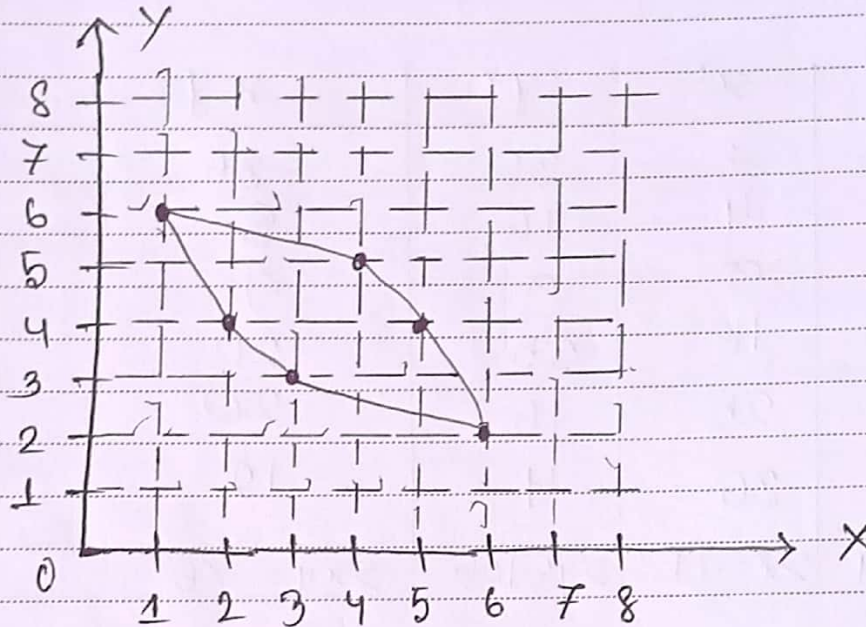






Comment: - Here  $r_{xy} = -0.68$ , so there exist negative correlation coefficient between  $x$  and  $y$

(2)



Scatter diagram —

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\* Regression: Regression is the functional relationship between two variable and of the two variable one may represent cause and the other may represent effect

$$y = a + bx$$

$\downarrow$  Dependent variable       $\rightarrow$  Regression coefficient  
 $\downarrow$  Constant/Intercept term       $\rightarrow$  Independent variable

\* Regression coefficient: - The mathematical measures of regression are called the coefficient of regression.

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

$$\therefore r_{xy} = \frac{n \sum xy - \sum x \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$$

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### \* Properties of regression coefficient :-

- ① Regression coefficient lies between  $-\infty$  to  $\infty$
- ② Regression coefficient is not symmetric i.e.  $b_{yx} \neq b_{xy}$
- ③ Correlation coefficient is the geometric mean of regression coefficient ;  $r_{xy} = \sqrt{b_{yx} \times b_{xy}}$
- ④ The arithmetic mean of regression coefficient is greater than correlation coefficient.

$$\left( \frac{b_{yx} + b_{xy}}{2} \right) \geq r_{xy}$$

- ⑤ It is not unit free.

### \* Uses of regression :-

- ① Whether a relationship exists or not.
- ② To find the strength of relationship
- ③ Determination of mathematical eq<sup>n</sup>
- ④ Prediction the value of dependent variable.

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\* Regression equation :-

$$y = a + bx$$

$$\Rightarrow a = y - bx$$

$$\Rightarrow \sum a = \sum y - b \sum x$$

$$\Rightarrow \frac{\sum a}{n} = \frac{\sum y}{n} - \frac{b \sum x}{n}$$

$$\Rightarrow \frac{a \cdot \sum 1}{n} = \bar{y} - b \bar{x} ; \left[ \text{Arithmetic, } \bar{x} = \frac{\sum x}{n} \right]$$

$$\Rightarrow \frac{a \cdot n}{n} = \bar{y} - b \bar{x}$$

$$\Rightarrow a = \bar{y} - b \bar{x} \therefore b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

Similarly,

$$x = a + by$$

$$\left( a = \bar{x} - b \bar{y} \therefore b = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2} \right)$$

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## \* Difference between regression & correlation coefficient :

| Correlation Coefficient                                                                                   | Regression Coefficient                                                   |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| (1) Numerical value by which we measure the strength of linear relationship between two or more variable. | Mathematical measure of regression are called coefficient of regression. |
| (2) Independent of change of origin and scale of measurement                                              | Independent of change of origin but not of scale.                        |
| (3) Lies between $-1$ to $+1$                                                                             | Lies between $-\infty$ to $\infty$                                       |
| (4) Symmetric ; $r_{xy} = r_{yx}$                                                                         | Not symmetric ; $b_{xy} \neq b_{yx}$                                     |
| (5) Unit free                                                                                             | Not unit free                                                            |
| (6) When $r=0$ , variable are correlated                                                                  | When $r=0$ then two line of regression are perpendicular to each other.  |







Qna: Consider the data,  $n = 6$

| $x$ | $y$ | $x^2$ | $y^2$ | $xy$ |
|-----|-----|-------|-------|------|
| 1   | 6   | 1     | 36    | 6    |
| 2   | 4   | 4     | 16    | 8    |
| 3   | 3   | 9     | 9     | 9    |
| 4   | 5   | 16    | 25    | 20   |
| 5   | 4   | 25    | 16    | 20   |
| 6   | 2   | 36    | 4     | 12   |

$$\Sigma x = 21 \quad \Sigma y = 24 \quad \Sigma x^2 = 91 \quad \Sigma y^2 = 106 \quad \Sigma xy = 75$$

- ① Compute the regression line of  $y$  on  $x$
- ② Predict/Estimate value of  $y$  when  $x = 4.5$

Soln

- ① The regression line of  $y$  on  $x$  is  

$$y = a + bx \quad \text{--- (i)}$$

$$\begin{aligned} \text{Here } a &= \bar{y} - b\bar{x} = \frac{\Sigma y}{n} - b \frac{\Sigma x}{n} \\ &= \frac{24}{6} - b \frac{21}{6} \quad \text{--- (ii)} \end{aligned}$$

$$\text{And } b = \frac{n \Sigma xy - \Sigma x \Sigma y}{n \Sigma x^2 - (\Sigma x)^2} \rightarrow -0.514$$





from (ii)

$$a = \frac{24}{6} - (-0.514) \frac{21}{6} = 5.8$$

from (i)

$$y = 5.8 - 0.514x$$

Therefore regression line of  $y$  on  $x$  is

$$y = 5.8 - 0.514x$$

Ans

(2) If  $x = 4.5$

$$y = 5.8 - 0.514(4.5)$$
$$= 3.487$$

Ans

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## Fundamentals of Probability

### \* Basic concept of probability :-

① Experiment :- It is an act that repeated under given condition.

Ex: Tossing coin, throwing a dice

② Outcome :- ~~Result~~ Result of an experiment.

③ Random experiment: Experiment are called random if the outcomes depend on chance and cannot be predicted with certainty.

Ex: Tossing of a fair coin, throwing of dice

④ Sample space :- Totality of all possible outcome of a random experiment.

Ex: Tossing a coin, we can have  $S = \{H, T\}$

Tossing two coin,  $S = \{HH, HT, TH, TT\}$

$[2^n]$

↳ each element called sample point.







⑤ Event: Any subset of a sample space.

Ex:  $S = \{HH, HT, TH, TT\}$

Event =  $\{HH\}$  or  $\{HH, TH\}$

⑥ Mutually exclusive event: Two event A and B said to be mutually exclusive if they have no common points.

Ex:  $S = \{1, 2, 3, 4, 5, 6\}$  dice

$A = \{1, 3, 5\}$      $B = \{2, 4, 6\}$

$A \cap B = \emptyset$

⑦ Non-Mutually exclusive event: Two event A and B said to be non-mutually exclusive if they have common points.

Ex:  $S = \{1, 2, 3, 4, 5, 6\}$  dice

$A = \{1, 3, 5\}$  ,  $B = \{2, 3, 4, 6\}$

$A \cap B = 3$







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⑧ Conditional probability :- If A and B are two event of any sample space 'S'. then the conditional probability of A for given value of B, denoted by

$$P[A|B] = \frac{P(AB)}{P(B)} ; P(B) > 0 \text{ or } B \text{ given } A$$

$$P[B|A] = \frac{P(AB)}{P(A)} ; P(A) > 0$$

$$P[B|A] \rightarrow \text{যদি A হলে}$$

$\rightarrow$  যখন A ঘটবে তখন B-এর সম্ভাব্যতা নির্ণয়।

⑨ Independent event :- Two event A and B are said to be independent if and only if one of the following condition holds:

$$(i) P[AB] = P[A]P[B]$$

$$(i) P[AB] \neq P[A]P[B]$$

$$(ii) P[A|B] = P[A]$$

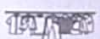
$$(ii) P[A|B] \neq P[A]$$

$$(iii) P[B|A] = P[B]$$

$$(iii) P[B|A] \neq P[B]$$

⑩ Dependent event :- if and only if one of the following condition holds:

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~~\* Classical or mathematical or a priori definition of probability:~~

\* Definition of Probability :-

① Classical or mathematical definition :- (given by Laplace)

Let  $n$  be the number of mutually exclusive, equally likely and exhaustive outcome of a random experiment and  $m$  of these events are favorable to an event  $A$ , then the prob<sup>y</sup> of  $A$  is denoted by

$$P(A) = \frac{\text{number of favorable number/outcomes}}{\text{number of total outcomes}}$$

$$= \frac{N(A)}{N(S)} = \frac{m}{n} ; 0 \leq P(A) \leq 1$$

② Axiomatic definition :-

let  $A$  be an event in any sample space  $S$ , then the prob<sup>y</sup> of  $A$  which is denoted by  $P(A)$  must satisfy the following axioms:

i)  $P(A) \geq 0$

ii)  $P(S) = 1$

iii)  $A_1$  &  $A_2$  are two mutually exclusive events, then

$$P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

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(3) Empirical or statistical probability :-

(4) Subjective probability :-

\* Law of probability :-

(i) Additive / Addition law of probab :-

$\Rightarrow$  If A and B two events, then

(1)  $P[A \cup B] = P[A] + P[B] - P[AB]$  ; Not mutually exclusive

(2)  $P[A \cup B] = P[A] + P[B]$  ; Mutually exclusive ( $P[AB] = 0$ )

$\Rightarrow$  If A, B and C are three events then,

(1) Non-mutually exclusive

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(\overset{AC}{\cancel{AB}}) - P(BC) + P(ABC)$$

(2) Mutually exclusive

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

(ii) Multiplication Law of probab :

$$P(AB) = P(B) P(A|B) = P(A) \cdot P(B|A)$$





\* The value of prob<sup>y</sup> lies between 0 and 1 : —

Proof let  $A$  be an event in  $S$ . Also let  
 $N(A) = m$ ,  $N(S) = n$ .

by defn  $P(A) = N(A)/N(S) = \frac{m}{n}$

It is clear that,  $0 \leq m \leq n$

$$\Rightarrow \frac{0}{n} \leq \frac{m}{n} \leq n/n \text{ [divided by } n]$$

$$\Rightarrow 0 \leq P(A) \leq 1 \text{ (Proved)}$$

\*  $P(A) + P(\bar{A}) = 1$

Proof From the complementary laws,  
 $A \cup \bar{A} = S$

Therefore,  $P(A \cup \bar{A}) = P(S)$

$$\Rightarrow P(A) + P(\bar{A}) = P(S) \text{ [} A, \bar{A} \text{ are mutually exclusive]}$$

$$\Rightarrow P(A) + P(\bar{A}) = 1 \text{ [axiomatic defn } P(S) = 1]$$

(Proved)

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\* Statement of Bayes theorem :-

Let  $B_1, B_2, \dots, B_n$  be  $n$  mutually exclusive and exhaustive events in a random experiment and  $D$  be any event in  $S$ . Then Bayes theorem states that,

$$P[B_i | D] = \frac{P[B_i] \cdot P[D | B_i]}{\sum_{j=1}^n P[B_j] \cdot P[D | B_j]} ; i = 1, 2, 3, \dots, n.$$

# In a factory, machines A, B, C manufacture respectively 25%, 35%, 40% of the product. Of total of their output 5, 4 and 2 percent are defective bolts respectively. A bolt is drawn at random from the product and is found to be defective.

So, What are the probab<sup>n</sup>s that it was manufactured by machine (i) A and (ii) C.

Sol<sup>n</sup>:  $B_1$ : An item produced by A ;  $P(B_1) = 0.25$   
 $B_2$ : An item produced by B ;  $P(B_2) = 0.35$   
 $B_3$ : " " " " by C ;  $P(B_3) = 0.4$   
 $D$ : Defective item produced by machines.





$$\therefore P(D|B_1) = 5\% = 0.05$$

$$P(D|B_2) = 4\% = 0.04$$

$$P(D|B_3) = 2\% = 0.02$$

$$\therefore P(B_i|D) = \frac{P(B_i) \cdot P(D|B_i)}{\sum_{j=1}^n P(B_j) \cdot P(D|B_j)}$$

$$P(B_1|D) = \frac{P(B_1) \cdot P(D|B_1)}{P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3)}$$
$$= 0.3623$$

$$P(B_3|D) = \frac{P(B_3) \cdot P(D|B_3)}{P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3)}$$
$$=$$

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## Random Variable

### Mathematical Expectation

\* Random Variable :- A random variable is a real valued function whose values are determined with the outcome of random experiment. (R.V)

Ex

|    | Head(s) |   |
|----|---------|---|
| HH | —       | 2 |
| HT | —       | 1 |
| TH | —       | 1 |
| TT | —       | 0 |

let  $X$  denotes the number of Heads, so  
 $X = 0, 1, 2$  Therefore  $X$  is The random variable.

\* Discrete R.V :- A random variable is called discrete if it can take only isolated values.

Ex: family members, phone numbers etc.

\* Continuous R.V :- A random variable is called continuous if it can take any value within certain limit.

Ex: Age, height, weight etc.





\* Probability function: A function  $f(x)$  of a discrete R.V "X" is called prob<sup>y</sup> function if it satisfies the Condition —

$$(i) f(x) \geq 0 \quad (ii) \sum_{x=0}^{\infty} f(x) = 1$$

\* Probability density function: A function  $f(x)$  of a continuous R.V "X" is called prob<sup>y</sup> density function if it satisfies the condition —

$$(i) f(x) \geq 0 \quad (ii) \int_{-\infty}^{\infty} f(x) dx = 1$$

\* Mathematical Expectation: — Let "X" be a discrete or continuous random variable (R.V) with probability or prob<sup>y</sup> density function  $f(x)$ , then the mathematical expectation of "X" is denoted by  $E[X]$  or  $\mu$

$$E[X] = \mu = \sum x f(x); \text{ if "X" is discrete}$$

$$= \int_{-\infty}^{\infty} x f(x) dx; \text{ if "X" is continuous.}$$





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\* Properties of Mathematical Expectation of a R.V.:

- ① If "a" is a constant, then  $E[a] = a$
- ② If "x" is a R.V. with expectation  $E[x]$ , then  $E[ax+b] = aE[x] + b$ ; a & b are constant
- ③ If "x" is a R.V. with M.E,  $E[x]$ , then  $E[x - E(x)] = 0$
- ④ If "x" & "y" are R.V., then  $E[x+y] = E[x] + E[y]$
- ⑤  $E[x-y] = E[x] - E[y]$
- ⑥  $E[a, x] = aE[x]$ ; a is a constant.

~~\* Properties~~\* Properties of variable of R.V.:

$$[S.D[x] = \sqrt{V[x]}]$$

- ①  $V[b] = 0$ ; b is a constant
- ②  $V[ax+b] = a^2 V[x]$ ; a & b are constant
- ③  $V[x] = E[x - E(x)]^2 = E[x]^2 - [E(x)]^2$ ; x = R.V
- ④  $V[x+y] = V[x] + V[y]$
- ⑤  $V[x-y] = V[x] + V[y]$
- ⑥  $V[ax] = a^2 V[x]$ ; a is constant.

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\* Problem:

Discrete R.V

|      |      |      |      |      |      |
|------|------|------|------|------|------|
| X    | 0    | 1    | 2    | 3    | 4    |
| f(x) | 0.10 | 0.20 | 0.20 | 0.25 | 0.15 |

i)  $P[X=1] = 0.20$

ii)  $P[X > 3] = P[X=4] = 0.15$

iii)  $P[1 < X < 4] = P[X=2] + P[X=3]$   
 $= 0.20 + 0.25 = 0.45$

iv)  $P[X < 1] = P[X=0] = 0.10$

Continuous R.V

$$f(x) = 3/4 (2x - x^2) ; 0 \leq x \leq 2$$

i)  $P[X > 1] = \int_1^2 f(x) dx$   
 $= \frac{3}{4} \int_1^2 (2x - x^2) dx$

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$$\begin{aligned}
 &= \frac{3}{4} \left\{ 2 \left[ \frac{x^2}{2} \right]_1^2 - \left[ \frac{x^3}{3} \right]_1^2 \right\} \\
 &= \frac{3}{4} \left\{ (2^2 - 1^2) - \frac{1}{3} (2^3 - 1^3) \right\} \\
 &= \frac{3}{4} \left( 3 - \frac{7}{3} \right) = \frac{1}{2} \quad \underline{\underline{\text{Ans}}}
 \end{aligned}$$

(ii)  $P(1.5 < x < 2.25)$

$$= \int_{1.5}^2 f(x) dx$$

$$= \frac{3}{4} \left\{ (2^2 - 1.5^2) - \frac{1}{3} (2^3 - 1.5^3) \right\}$$

$$= \frac{3}{4} \times \frac{5}{24} = \frac{5}{32} \quad \underline{\underline{\text{Ans}}}$$

✱

| X    | -1  | 0   | 1    | 2    | 3   |
|------|-----|-----|------|------|-----|
| f(x) | 0.2 | 0.1 | 0.25 | 0.15 | 0.3 |

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$$\textcircled{i} P[X \leq 1] = P[X=0] + P[X=-1]$$

$$= 0.1 + 0.2 = 0.3 \quad \underline{\underline{\text{Ans}}}$$

$$\textcircled{ii} P[-1 < X < 2] = P[X=0] + P[X=1] = 0.35 \quad \underline{\underline{\text{Ans}}}$$

$$\textcircled{iii} P[X > 0] = P[X=1] + P[X=2] + P[X=3]$$

$$= 0.7 \quad \underline{\underline{\text{Ans}}}$$

$$\textcircled{iv} E[X] = \sum x f(x)$$

$$= (-1 \times 0.2) + (0 \times 0.1) + (1 \times 0.25) + (2 \times 0.15) + (3 \times 0.3)$$

$$= 1.25 \quad \underline{\underline{\text{Ans}}}$$

$$\textcircled{v} E[X^2] = \sum x^2 f(x)$$

$$= (1 \times 0.2) + (0 \times 0.1) + (1 \times 0.25) + (4 \times 0.15) + (9 \times 0.3)$$

$$= 3.75$$

$$\therefore V[X] = E[X^2] - [E(X)]^2$$

$$= 3.75 - (1.25)^2 = 2.1875$$

$$\therefore S.D[X] = \sqrt{V[X]} = 1.48 \quad \underline{\underline{\text{Ans}}}$$







#  $f(x) = kx^y ; 0 \leq x \leq 1$

- (i) the value of  $k$
- (ii)  $X$  lies between 0.2 and 0.3
- (iii)  $X$  lies than 0.5
- (iv)  $X$  greater than 0.75
- (v) Mean, variance, SD and CV
- (vi)  $V[7x+9]$

Soln:

$$(i) \int_{-\infty}^{\infty} f(x) dx = 1$$

$$= \int_{-\infty}^{\infty} kx^y dx = 1$$

$$= k \left[ \frac{x^{y+1}}{y+1} \right]_{-\infty}^{\infty} = 1$$

$$= k/3 = 1 \Rightarrow k = 3$$

$$\therefore f(x) = 3x^y \text{ Ans}$$

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(ii)  $P[0.2 < x < 0.3]$

$$= \int_{0.2}^{0.3} 3x^2 dx$$

$$= [0.3^3 - 0.2^3] = 0.027 - 0.008 = 0.019 \quad \underline{\underline{Ans}}$$

(iii)  $P[x < 0.5]$

$$= \int_0^{0.5} f(x) dx = [0.5^3 - 0^3] = 0.125 \quad \underline{\underline{Ans}}$$

(iv)  $P[x > 0.75]$

$$\Rightarrow \int_{0.75}^1 f(x) dx = [1^3 - 0.75^3] = 0.578 \quad \underline{\underline{Ans}}$$

(v)

$$\text{Mean} = \mu = E[x] = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^1 x \cdot 3x^2 dx = 3 \int_0^1 x^3 dx = 3 \left[ \frac{x^4}{4} \right]_0^1 = 3/4 \quad \underline{\underline{Ans}}$$

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Date:

$$V[X] = E[X^2] - [E(X)]^2$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= 2 \int_0^1 x^4 dx = 2/5$$

$$\therefore V[X] = 2/5 - (2/4)^2 = 2/8 \quad \underline{\underline{Ans}}$$

$$SD[X] = \sqrt{V(X)} = \sqrt{2/8} \quad \underline{\underline{Ans}}$$

$$C.V = \frac{SD}{\text{mean}} \times 100 =$$

$$(vi) V[7x+9] = 7^2 V[X] + 0 \quad \underline{\underline{Ans}}$$

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## Probability Distribution

- (i) Binomial distribution
  - (ii) Poisson distribution
  - (iii) Normal distribution
- Discrete
- Continuous

\* Binomial distribution :- (नियत) A discrete R.V is said to be binomial dist<sup>n</sup> if prob<sup>y</sup> function is as follow -

$$P(x) = {}^nC_x p^x q^{n-x}; x = 0, 1, 2, \dots, n$$

[ $n \leq 20$ ] If  $n > 20$ , it goes to poisson dist<sup>n</sup>

where  $n$  and  $p$  are the parameter of the dist<sup>n</sup> and  $p + q = 1$ .

Here,  $n$  = number of trial.

$x$  = number of success.

$p$  = prob<sup>y</sup> of success.

$q$  = prob<sup>y</sup> of failure.

Note: Parameter is the unknown constant of the population.

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### \* Assumption/Properties of binomial dist<sup>n</sup>:

- (1) The number of trials  $n$  is fixed
- (2) Trials are independent.
- (3) Every trials has two possible "success"/"failure".
- (4) Prob<sup>y</sup> of success & prob<sup>y</sup> of failure remain same from trial to trail

### \* Problem:

A family has five/5 children, the prob<sup>y</sup> of being a boy or a girl equal. Find the prob<sup>y</sup> that

- (i) No boy, boys
- (ii) Exactly 2 boys
- (iii) At most 2 boys
- (iv) At least 3 boys.

Soln:

Here,  $n = 5$ ,  $p = q = 0.5 / 1/2$

$$\therefore P(x) = {}^5C_x (0.5)^x (0.5)^{5-x} \quad \text{--- (1)}$$

(i) No boys,

$$P[x=0] = P(0) = {}^5C_0 (0.5)^0 (0.5)^{5-0} = 0.03125$$

Ans







(ii) Exactly 2 boys -

$$P[X=2] = P(2) = {}^5C_2 (0.5)^2 (0.5)^{5-2}$$

=

(iii) At most 2 boys -

$$P[X \leq 2] = P[X=0] + P[X=1] + P[X=2]$$

$$= {}^5C_0 (0.5)^0 (0.5)^{5-0} + {}^5C_1 (0.5)^1 (0.5)^{5-1} + {}^5C_2 (0.5)^2 (0.5)^{5-2}$$

$$= 0.5 \text{ Ans}$$

(iv) At least 3 boys -

$$P[X \geq 3] = P[X=3] + P[X=4] + P[X=5]$$

$$= 0.5 \text{ Ans}$$

H.W : A fair coin is tossed 5 times. find the prob<sup>y</sup> (i) Exactly 2 heads (ii) at least 3 heads (iii) No heads (iv) at most 2 heads.







Date:

\* Poisson Distribution:- A discrete R.V is said to have poisson distribution if its prob<sup>y</sup> function is as

$$f(x) = \frac{e^{-m} m^x}{x!}; x = 0, 1, 2, 3, \dots \infty$$

where  $m$  is the parameter &  $e = 2.718$

[ $n \uparrow \rightarrow P \downarrow$ ]

\* Normal distribution:- A continuous R.V is said to have normal dist<sup>n</sup> if its prob<sup>y</sup> density function is

$$f(x, \mu, \sigma^2) = \frac{e^{-1/2 \left( \frac{x-\mu}{\sigma} \right)^2}}{\sigma \sqrt{2\pi}}; \begin{matrix} -\infty < x < \infty \\ -\infty < \mu < \infty \\ \sigma^2 \geq 0 \end{matrix}$$

where  $\mu$  and  $\sigma^2$  are the parameters of dist<sup>n</sup>

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## Estimation & test of hypothesis

(\*) Hypothesis :- A statement about the nature of a population.

Ex: Students who eat breakfast will perform better on a ~~stat~~ exam than students who do not eat breakfast.

(\*) Test of hypothesis :- The statistical procedure which is used to verify any statement/assumption about population parameters on the basis of sample observation.

(\*) NULL hypothesis :- The hypothesis which we are going to test for possible rejection under the assumption that it is true.

denoted by :  $H_0$  ;  $H_0 : \mu_1 = \mu_2$   $\rightarrow$  Mean / s/y

(\*) Alternative hypothesis :- Each of all possible hypothesis other than null hypothesis is called alternative hypothesis.

denoted by  $H_1$  ;  $H_A : \mu_1 \neq \mu_2$



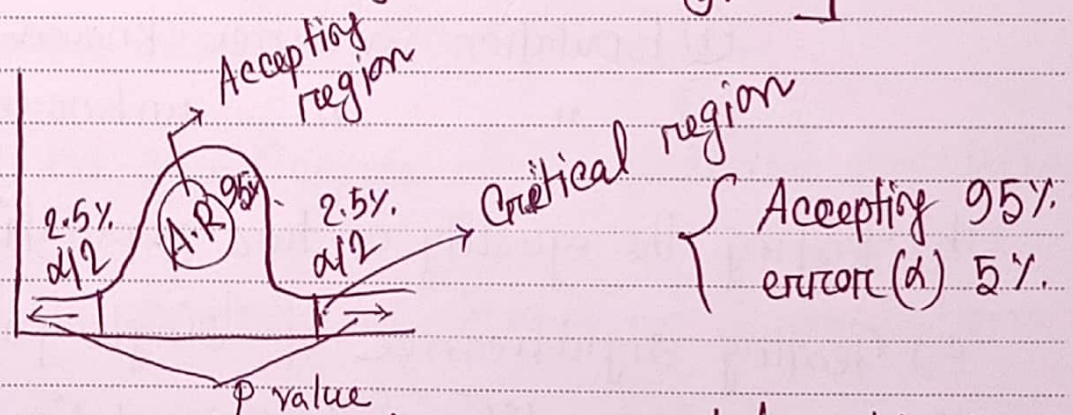


\* Type-I-error: (false positive) :- The error of rejecting  $H_0$  (accepting  $H_1$ ) which is true is called of first kind error.

\* Type-II error: (false negative) :- The error of accepting null hypothesis  $H_0$  when it is false is called error of 2<sup>nd</sup> kind. denoted by  $\beta$

\* level of significance :- ( $\alpha$ ) The prob<sup>y</sup> of type-I error denoted by " $\alpha$ " is called level of significance.

[ P-value : It is the least possible value of  $\alpha$  for which we can reject the null hypoth ]



$\Rightarrow$  If  $p < 1\%/0.01$ , the result are regarded as highly significant

$\Rightarrow$  If  $p < 5\%/0.05$ , The result are regarded as statistically significant

$\Rightarrow$  If  $p > 10\%/0.10$ , considered not significant.

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(\*) Commonly used test statistic:

- ① ~~Q-Q~~ The normal test (Z-test)
- ② The t test
- ③ Chi-square ( $\chi^2$ ) test
- ④ F-test

When we used this?

(#) The normal test :-

a) Testing the significance of population mean

① Population variance known

② " " unknown

b) Testing the equality of two population mean.

c) Testing significance of single population.

d) " the equality of two population proportions.







### #1) The t test :-

- To test the significance of population mean  $\mu$ .
- " " " in difference between two population means.
- " " " ~~equal~~ of population co-relation coefficient and regression co-efficient
- " " the equality of two co-related means.
- " " " of two independent regression co-efficient.

### #2) Chi-square test :- [For exam]

- To test the significance of a specified population variance
- " " the goodness of fit of a distribution.
- " " the independence of attributes.
- " " the homogeneity of several test.
- " " " of several population variance.







(#) F-test :-

- a) To test the equality of several population means
- b) " " " two population variance
- c) " " the significance of regression
- d) " " " of multiple co-relation co-efficient.

Q: Two hundred engineers were interviewed and classified according to their result and job satisfaction. The distribution of graduates by result and job satisfaction are given.

| Results   | Job satisfaction |    |
|-----------|------------------|----|
|           | Yes              | NO |
| Excellent | 20               | 70 |
| Good      | 45               | 65 |

Compute the value of chi-square of above data.





Date:

Soln:

| Results   | Job Satisfaction |                 | Total         |
|-----------|------------------|-----------------|---------------|
|           | Yes              | No              |               |
| Excellent | 20 ( $O_{11}$ )  | 70 ( $O_{12}$ ) | 90 ( $R_1$ )  |
| Good      | 45 ( $O_{21}$ )  | 65 ( $O_{22}$ ) | 110 ( $R_2$ ) |
| Total     | 65 ( $C_1$ )     | 135 ( $C_2$ )   | 200 = N       |

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$\chi^2$  chi-square

O = Observed value  
E = Expected value

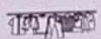
Here,  $E_{ij} = \text{Row}_i \times \text{Col}_j / N$

$$E_{11} = \frac{90 \times 65}{200} = 29.25$$

$$E_{12} = \frac{90 \times 135}{200} = 60.75$$

$$E_{21} = 35.75 \quad E_{22} = 74.25$$

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Therefore,

$$\chi^2 = \frac{(20 - 29.25)^2}{29.25} + \frac{(70 - 60.75)^2}{60.75} + \frac{(45 - 35.75)^2}{35.75} +$$

$$\frac{(65 - 74.25)^2}{74.25}$$

$$= 2.93 + 1.41 + 2.39 + 1.15$$

$$= 6.88$$

\* If Qns asked for giving comment then, we have to compare it with Table value of chi square.

So, if  $\chi^2 > \chi^2_{\text{table}}$ : Accepted

otherwise Not Accepted.

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# Critical values of the Chi-square distribution with $d$ degrees of freedom

| Probability of exceeding the critical value |        |        |        |     |        |        |        |
|---------------------------------------------|--------|--------|--------|-----|--------|--------|--------|
| $d$                                         | 0.05   | 0.01   | 0.001  | $d$ | 0.05   | 0.01   | 0.001  |
| 1                                           | 3.841  | 6.635  | 10.828 | 11  | 19.675 | 24.725 | 31.264 |
| 2                                           | 5.991  | 9.210  | 13.816 | 12  | 21.026 | 26.217 | 32.910 |
| 3                                           | 7.815  | 11.345 | 16.266 | 13  | 22.362 | 27.688 | 34.528 |
| 4                                           | 9.488  | 13.277 | 18.467 | 14  | 23.685 | 29.141 | 36.123 |
| 5                                           | 11.070 | 15.086 | 20.515 | 15  | 24.996 | 30.578 | 37.697 |
| 6                                           | 12.592 | 16.812 | 22.458 | 16  | 26.296 | 32.000 | 39.252 |
| 7                                           | 14.067 | 18.475 | 24.322 | 17  | 27.587 | 33.409 | 40.790 |
| 8                                           | 15.507 | 20.090 | 26.125 | 18  | 28.869 | 34.805 | 42.312 |
| 9                                           | 16.919 | 21.666 | 27.877 | 19  | 30.144 | 36.191 | 43.820 |
| 10                                          | 18.307 | 23.209 | 29.588 | 20  | 31.410 | 37.566 | 45.315 |

INTRODUCTION TO POPULATION GENETICS, Table D.1



1. Following figures give the rainfall in inches for 7 districts and the production of rabi crop in hundred kgs per acre of a particular year.

|                  |    |    |    |    |    |    |    |
|------------------|----|----|----|----|----|----|----|
| Rainfall:        | 20 | 22 | 24 | 26 | 28 | 30 | 32 |
| Rabi Production: | 30 | 35 | 40 | 50 | 60 | 62 | 55 |

- Compute coefficient of correlation between the rainfall and production and comment
  - Draw Scatter Diagram
  - Obtain the regression line of y on x.
  - Estimate the total rabi crop (kgs) when rainfall is 35 inches
  - Obtain the regression line of x on y.
2. A computer center has 100 computers which are collected from three companies A, B, and C. The selected computers from these companies are 50, 30 and 20 respectively. The probabilities of trouble which is faced in these computers daily are 0.15, 0.20, and 0.25 respectively. One day during work a computer is found defective. What is the probability that it was collected from company B?
3. A continuous random variable X has the following probability density function:  

$$f(x) = P(2x - x^2); 0 < x < 2$$
 Compute (i) the value of P (ii) E [X], (iii) S.D [X] (iv) E[3x + 6] (v) V[4x + 9] (vi) P(X > 1) And P(X < 0.75)
4. Ten friends take a picnic to a cricket match. As his contribution to the picnic, Salim buys ten sandwiches at a supermarket. He selects the sandwiches at random from those on display. The probability that a sandwich is suitable for vegetarian is independently 0.4 for each sandwich. Find the probability that, of the ten sandwiches, the number suitable for vegetarian is: (i) 4 or less; (ii) exactly 5; (iii) At least 2.
5. The following contingency table shows the classification of 200 peoples according to the gender and infected with the COVID-19:

| COVID-19 | Gender |        |
|----------|--------|--------|
|          | Male   | Female |
| Positive | 60     | 30     |
| Negative | 75     | 35     |

Test whether there is any relationship between infected with the COVID-19 and gender preference at  $\alpha=0.05$ . At 5% level of significance tabulated value of Chi-square @ 1df = 3.84



Ans to the Q NO. 01

|              |     |      |      |      |      |      |      |
|--------------|-----|------|------|------|------|------|------|
| Rainfall (x) | 20  | 22   | 24   | 26   | 28   | 30   | 32   |
| Rabi (y)     | 30  | 35   | 40   | 50   | 60   | 62   | 55   |
| $x^2$        | 400 | 484  | 576  | 676  | 784  | 900  | 1024 |
| $y^2$        | 900 | 1225 | 1600 | 2500 | 3600 | 3844 | 3025 |
| $xy$         | 600 | 770  | 960  | 1300 | 1680 | 1860 | 1760 |

Here,  $n = 7$ ,

$$\sum x = 182, \quad \sum y = 332$$

$$\sum x^2 = 4844, \quad \sum y^2 = 16694$$

$$\sum xy = 8930$$

(i) We know that,

$$r_{xy} = \frac{n \sum xy - \sum x \cdot \sum y}{\sqrt{n(\sum x^2) - (\sum x)^2} \cdot \sqrt{n(\sum y^2) - (\sum y)^2}}$$

$$= \frac{(7 \times 8930) - (182 \times 332)}{\sqrt{(7 \times 4844) - 33124} \cdot \sqrt{(7 \times 16694) - 110224}}$$

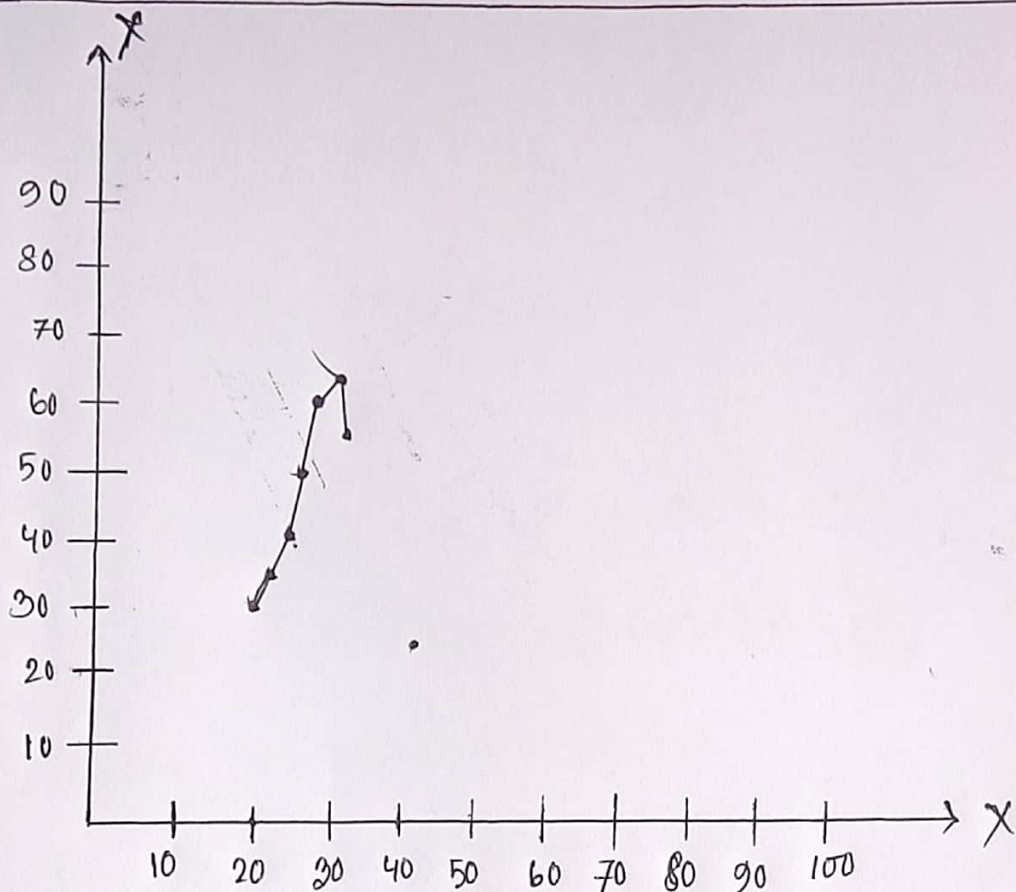
$$= \frac{62510 - 60424}{\sqrt{784} \cdot \sqrt{6634}}$$

$$= \frac{2086}{28 \cdot 81.449} = 0.9147$$

Here,  $r_{xy} = 0.9147$ . So there exist strongly positive correlation coefficient.



(ii)



(iii) The regression line of  $y$  on  $x$   
 $y = a + bx$  — (1)

$$\text{Here, } a = \bar{y} - b\bar{x} = \frac{\sum y}{n} - b \frac{\sum x}{n}$$
$$= \frac{332}{7} - b \frac{182}{7} \quad \text{--- (2)}$$

$$\text{And } b = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum x^2 - (\sum x)^2}$$
$$= \frac{7 \times 8930 - 60424}{784}$$
$$= 2.6607$$



from (2)

$$\begin{aligned} a &= \frac{332}{7} - (2.6607) \frac{182}{7} \\ &= 47.429 - 69.1782 \\ &= -21.7492 \end{aligned}$$

Therefore, regression line of  $y$  on  $x$  is

$$y = -21.7492 + (2.6607)x \quad \underline{\underline{\text{Ans}}}$$

(iv) When,  $x = \text{rainfall} = 35$

$$\begin{aligned} y &= -21.7492 + (2.6607)(35) \\ &= 71.38 \quad \underline{\underline{\text{Ans}}} \end{aligned}$$

(v) The regression line of  $x$  on  $y$

$$x = a + by \quad \text{--- (1)}$$

$$\begin{aligned} \text{Here, } a = x - by &= \frac{\sum x}{n} - b \frac{\sum y}{n} \\ &= \frac{182}{7} - b \frac{332}{7} \quad \text{--- (2)} \end{aligned}$$

$$b = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum y^2 - (\sum y)^2}$$

$$= \frac{2086}{6624} = 0.31445$$



from (2)

$$\begin{aligned} a &= \frac{182}{7} - (0.31445) \frac{332}{7} \\ &= 26 - 14.9139 \\ &= 11.086 \end{aligned}$$

Therefore regression line of  $x$  on  $y$

$$x = 11.086 + (0.31445) y \quad \underline{\underline{Ans}}$$

Ans to the Q NO: 02

$B_1$ : The computer collected from A ;  $P(B_1) = 50$

$B_2$ : The computer collected from B ;  $P(B_2) = 30$

$B_3$ : The computer collected from C ;  $P(B_3) = 20$

D : Defective computer.

$$P(D|B_1) = 0.15$$

$$P(D|B_2) = 0.20$$

$$P(D|B_3) = 0.25$$

$$\begin{aligned} \therefore P(B_2|D) &= \frac{P(B_2) \cdot P(D|B_2)}{P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3)} \\ &= \frac{30 \times 0.20}{(50 \times 0.15) + (30 \times 0.20) + (20 \times 0.25)} \\ &= 0.2243 \end{aligned}$$

Ans



Ans to the Q NO:03

$$f(x) = p(2x - x^2) ; 0 < x < 2$$

(i) let,  $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_{-\infty}^{\infty} p(2x - x^2) dx = 1$$

$$\Rightarrow p \left\{ 2 \int_0^2 x dx - \int_0^2 x^2 dx \right\} = 1$$

$$\Rightarrow p \left\{ 2 \cdot \frac{x^2}{2} \Big|_0^2 - \frac{x^3}{3} \Big|_0^2 \right\} = 1$$

$$\Rightarrow p \left( 4 - \frac{8}{3} \right) = 1$$

$$\Rightarrow p \left( \frac{2}{3} \right) = 1 \quad \Rightarrow p = \frac{3}{2}$$

$\therefore p = \frac{3}{2}$  Ans!

(ii)  $E[x] = \int_{-\infty}^{\infty} x f(x) dx$

$$= \int_0^2 x \cdot \frac{3}{2} (2x - x^2) dx$$

$$= \frac{3}{2} \int_0^2 (2x^2 - x^3) dx$$

$$= \frac{3}{2} \left\{ 2 \int_0^2 x^2 dx - \int_0^2 x^3 dx \right\}$$



$$\begin{aligned}
 &= \frac{2}{4} \left\{ 2 \left( \frac{x^3}{3} \right)_0^2 - \left( \frac{x^4}{4} \right)_0^2 \right\} \\
 &= \frac{2}{4} \left( 2 \times \frac{8}{3} - \frac{16}{4} \right) \\
 &= \frac{2}{4} \left( \frac{16}{3} - 4 \right) = \frac{2}{4} \times \frac{4}{3} \\
 &= 1
 \end{aligned}$$

$$\therefore E[X] = 1$$

$$\begin{aligned}
 \textcircled{\text{iii}} \quad E[X^2] &= \int_{-\infty}^{\infty} x^2 f(x) dx \\
 &= \int_0^2 x^2 \cdot \frac{2}{4} (2x - x^2) dx \\
 &= \frac{2}{4} \int_0^2 (2x^3 - x^4) dx \\
 &= \frac{2}{4} \left\{ 2 \cdot \frac{x^4}{4} \Big|_0^2 - \frac{x^5}{5} \Big|_0^2 \right\} \\
 &= \frac{2}{4} \left( \frac{16}{2} - \frac{32}{5} \right) \\
 &= \frac{2}{4} \cdot \frac{8}{5} = \frac{1 \times 2}{5} \cdot \frac{6}{5}
 \end{aligned}$$

$$\begin{aligned}
 \therefore V[X] &= E[X^2] - [E(X)]^2 \\
 &= \frac{6}{5} - 1 \\
 &= \frac{1}{5} = 0.2
 \end{aligned}$$



$$S.D [X] = \sqrt{V[X]} = \sqrt{0.2} = 0.447 \underline{\underline{Am'}}$$

$$\begin{aligned} \textcircled{iv} E[3x+6] &= 3E[X] + 6 \\ &= (3 \times 1) + 6 \\ &= 3+6 = 9 \underline{\underline{Am!}} \end{aligned}$$

$$\begin{aligned} \textcircled{v} V[4x+9] &= 16 V[X] + 0 \\ &= 16 \times 0.2 \\ &= 3.2 \underline{\underline{Am!}} \end{aligned}$$

$$\begin{aligned} \textcircled{vi} P(x > 1) &= \int_1^2 \frac{3}{4} (2x - x^2) dx \\ &= \frac{3}{4} \left\{ 2 \cdot \left( \frac{x^2}{2} \right)_1^2 - \left( \frac{x^3}{3} \right)_1^2 \right\} \\ &= \frac{3}{4} \left\{ (4-1) - \frac{1}{3}(8-1) \right\} \\ &= \frac{3}{4} \left( 3 - \frac{7}{3} \right) \\ &= 0.5 \underline{\underline{Am!}} \end{aligned}$$



$$P(X < 0.75)$$

$$= \int_0^{0.75} \frac{3}{4} (2x - x^2) dx$$

$$= \frac{3}{4} \left\{ (x^2)_0^{0.75} - \frac{1}{3} (x^3)_0^{0.75} \right\}$$

$$= \frac{3}{4} \left\{ \frac{9}{16} - \frac{9}{64} \right\}$$

$$= \frac{3}{4} \times \frac{27}{64}$$

$$= \frac{81}{256} \quad \underline{\underline{\text{Ans}}}$$

Ans to the Q NO: 04

Here,  $n = 10$ ,  $p = 0.4$ ,  $q = (1 - 0.4) = 0.6$

We know that,

$$f(x) = {}^{10}C_x (0.4)^x (0.6)^{10-x} \quad \text{--- (1)}$$

(i) 4 or less

$$P[X \leq 4] = P[X=0] + P[X=1] + P[X=2] + P[X=3] + P[X=4]$$

$$= \left\{ {}^{10}C_0 (0.4)^0 (0.6)^{10-0} \right\} + \left\{ {}^{10}C_1 (0.4)^1 (0.6)^{10-1} \right\} +$$

$$\left\{ {}^{10}C_2 (0.4)^2 (0.6)^{10-2} \right\} + \left\{ {}^{10}C_3 (0.4)^3 (0.6)^{10-3} \right\} + \left\{ {}^{10}C_4 (0.4)^4 (0.6)^{10-4} \right\}$$



$$= (1 \times 1 \times 6.0466 \times 10^{-3}) + (10 \times 0.4 \times 0.01) + (45 \times 0.16 \times 0.0168) \\ + (120 \times 0.064 \times 0.2799) + (210 \times 0.0256 \times 0.04666)$$

$$= (6.0466 \times 10^{-3}) + 0.04 + 0.12 + 0.21496 + 0.2508$$

$$= \cancel{0.56649} \quad \underline{\underline{Ans}} \quad 0.6318 \quad \underline{\underline{Ans}}$$

(ii) Exactly 5

$$P[X=5] = {}^{10}C_5 (0.4)^5 (0.6)^{10-5} \\ = 252 \times 0.01024 \times 0.07776 \\ = 0.20066$$

(iii) At least 2

$$P[X \geq 2] = P[X=2] + P[X=3] + P[X=4] + P[X=5] \\ + P[X=6] + P[X=7] + P[X=8] + P[X=9] \\ + P[X=10]$$

$$\Rightarrow 0.12 + 0.21496 + 0.2508 + 0.20066 + \\ {}^{10}C_6 (0.4)^6 (0.6)^{10-6} + {}^{10}C_7 (0.4)^7 (0.6)^{10-7} + {}^{10}C_8 (0.4)^8 (0.6)^{10-8} \\ + {}^{10}C_9 (0.4)^9 (0.6)^{10-9} + {}^{10}C_{10} (0.4)^{10} (0.6)^{10-10}$$

$$\Rightarrow \cancel{0.56649} + 0.1115 + 0.0425 + 0.0106 + (1.572 \times 10^{-3}) \\ + (1.0485 \times 10^{-4})$$

$$= 0.798 \quad \underline{\underline{Ans}}$$



Ans to the Q NO. 05

| COVID-19 | Gender          |                 | Total         |
|----------|-----------------|-----------------|---------------|
|          | Male            | Female          |               |
| Positive | 60 ( $O_{11}$ ) | 30 ( $O_{12}$ ) | 90 ( $R_1$ )  |
| Negative | 75 ( $O_{21}$ ) | 35 ( $O_{22}$ ) | 110 ( $R_2$ ) |
| Total    | 135 ( $C_1$ )   | 65 ( $C_2$ )    | 200 = N       |

We know that,

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad \text{--- (i)}$$

Here,

$$E_{11} = \frac{90 \times 135}{200} = 60.75$$

$$E_{12} = \frac{90 \times 65}{200} = 29.25$$

$$E_{21} = \frac{110 \times 135}{200} = 74.25$$

$$E_{22} = \frac{110 \times 65}{200} = 35.75$$

From eqn (i)

$$\chi^2 = \left\{ \frac{(60 - 60.75)^2}{60.75} + \frac{(30 - 29.25)^2}{29.25} + \frac{(75 - 74.25)^2}{74.25} + \frac{(35 - 35.75)^2}{35.75} \right\}$$



$$\chi^2 = \frac{1}{108} + \frac{1}{52} + \frac{1}{122} + \frac{9}{572}$$

$$= 0.0518$$

$\alpha = 0.05$  which is greater equal to the  $\chi^2$   
 again,  $1df = 3.84$  is greater than  $\chi^2$ , so  
 There is null hypothesis.

Therefore, There is no relationship between infected  
 with the COVID-19 and gender preference at  
 $\alpha = 5\%$  or at the 0.05 level of significance.

\_\_\_\_\_



## 1. Set up Hypotheses:

- Null Hypothesis ( $H_0$ ): There is no relationship between COVID-19 infection and gender.
- Alternative Hypothesis ( $H_1$ ): There is a relationship between COVID-19 infection and gender.



## 5. Compare with Critical Value:

- Compare the calculated chi-square statistic with the critical value at 1 degree of freedom ( $df=1$ ) and a significance level of 0.05.

## 6. Make a Decision:

- If the calculated chi-square statistic is greater than the critical value, reject the null hypothesis and conclude that there is a significant relationship between COVID-19 infection and gender.